MEC-102

MACROECONOMIC ANALYSIS

School of Social Sciences
Indira Gandhi National Open University
Maidan Garhi, New Delhi-110068

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COURSE PREPARATION TEAM

Block/ Unit Title		Unit Writer
Block 1	**Traditional Approaches to Macroeconomics**	
Unit 1	The Classical Approach	Dr. Jagannath Mallick, Independent Researcher, New Delhi
Unit 2	The Keynesian Model	Prof. Ananya Ghosh Dastidar, University of Delhi
Unit 3	Neoclassical Synthesis	Prof. Kaustuva Barik, IGNOU
Unit 4	Open Economy Macroecon. - I	Prof. Ananya Ghosh Dastidar, Department of Business and Finance, University of Delhi
Unit 5	Open Economy Macroecon. - II	
Block 2	**Expectations and Macroeconomics**	
Unit 6	Inflation and Unemployment	Prof. Kaustuva Barik, IGNOU
Unit 7	Rational Expectations	Dr. Manjula Singh, St. Stephens College, University of Delhi
Block 3	**Intertemporal Decision-Making**	
Unit 8	Consumption and Asset Prices	Prof. Mausumi Das, Delhi School of Economics, Delhi
Unit 9	Ramsey-Cass-Koopmans Model	
Unit 10	Overlapping Generations Model	
Block 4	**Theories of the Business Cycles**	
Unit 11	Traditional Models of Business Cycles	Dr. Archi Bhatia, Associate Professor, Ramjas College, University of Delhi
Unit 12	Real Business Cycles	Dr. Jagannath Mallick, Independent Researcher, Delhi and Prof. Kaustuva Barik, IGNOU
Block 5	**Labour Markets**	
Unit 13	Nominal and Real Rigidities	Prof. Avadhhot Nadkarni (retd.), University of Mumbai
Unit 14	Search Theory and Unemployment	
Block 6	**Issues in Monetary Policy**	
Unit 15	Central Banks and the Supply of Money	Dr. Jagannath Mallick, Independent Researcher, Delhi
Unit 16	Conduct of Monetary Policy	Prof. Kaustuva Barik and Ms. Apica Sharma, IGNOU
Unit 17	Theory of Monetary Policy	Dr. Vineet Kohli, Tata Institute of Social Sciences, Mumbai
Block 7	**Fiscal Policy and Macroeconomics**	
Unit 18	Fiscal Policy	Dr. Archi Bhatia, Associate Professor, Ramjas College, University of Delhi
Unit 19	Fiscal Sustainability	

Course Coordinator: Prof. Kaustuva Barik

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COURSE INTRODUCTION

Macroeconomics analyses and establishes functional relationships among economy level aggregates. Aggregative analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essential for proper comprehension of various issues and policies. Macroeconomics now is not only a scientific method of analysis but also a body of empirical economic knowledge.

After going through this course a student is expected to

- have an intuitive understanding of the theoretical developments in macroeconomics;
- contrast among various schools of macroeconomic thought;
- analyse real world issues with the help of macroeconomic theories; and
- apply macroeconomic theory to empirical data.

The course comprises seven blocks.

Block 1, entitled **Traditional Approaches to Macroeconomics**, is somewhat introductory in nature. After explaining some basic concepts that are used frequently in macroeconomics and form building blocks of the subject, it presents the classical and Keynesian views on determination of output and employment in an economy. Subsequently, it presents a synthesis of the classical and Keynesian ideas through the tools of IS-LM curves. The block also deals with issues pertaining to open economy macroeconomics.

Block 2, entitled **Expectations and Macroeconomics**, deals with issues such as inflation and unemployment. It traces the evolution of Phillips Curve – how it originated as a relationship between real and monetary variables; how it occupied a prime place as an analytical tool in Keynesian economics; and how it became irrelevant in the new-classical economics. In the course of discussion, this block deals with two important concepts, viz., adaptive expectations and rational expectations.

Block 3, entitled **Intertemporal Decision-making**, analyses the behavior of economic agents such as households and firms in an inter-temporal framework. It discusses the Ramsey-Cass-Koopmans model and overlapping-generations model. It also looks into issues such as the choice between work and leisure on the part of households.

Block 4, entitled **Theories of Business Cycles**, is devoted to the analysis of economic fluctuations in an economy, particularly the business cycles. It begins with traditional theories of business cycles and later moves on to real business cycles models with inter-temporal substitution and the propagation mechanism.

Block 5, entitled **Labour Markets**, discusses issues related to unemployment in the economy. Apart from presenting nominal and real rigidities, the block throws lights on search theories and unemployment.

The subject matter of **Block 6** is **Issues in Monetary Policy**. It deals with issues such as money supply, conduct of monetary policy, and theory of monetary policy. In recent years inflation targeting, policy rules, and dynamic consistency have been important concerns of policy makers.

Block 7, entitled **Fiscal Policy and Macroeconomics**, looks into the implications fiscal policy in an economy. Debt sustainability and Ricardian equivalence are two important issues discussed in this Block.

UNIT 1 THE CLASSICAL APPROACH*

Structure

1.1 OBJECTIVES

After going through the unit, you should be in a position to

- bring out the salient features of classical economics;
- analyse the classical approach to the determination output and employment;
- explain the quantity theory of money and its implication on price determination;
- explain the classical dichotomy in the economy and the neutrality of money in a classical system; and
- describe the behavior of prices in the long-run and short-run.

1.1 INTRODUCTION

Being a student of economics, you must be aware that macroeconomic theory has evolved over time in response to the dynamic macroeconomic environment. Initially, during the nineteenth and early twentieth century, there was a consensus among economists and economic historians that an economy could run smoothly without much fluctuation in income, output and employment. The perception

* Dr. Jagannath Mallick, Independent Researcher, New Delhi writer

was that there should be minimum state intervention in economic variables such as wages, prices and interest rates. The market forces (supply and demand) would take care of economic equilibrium in an economy. Such an assumption, however, is unrealistic – all governments have in place certain economic policies (such as monetary policy and fiscal policy) and they carry out amendments to these policies from time to time!

Economic theory refers to a set of ideas and principles which are used to explain, analyse and predict certain 'phenomenon' or observable events. As you know, macroeconomic theory deals with macroeconomic phenomena, which are aggregative in nature. The need for a special branch of macroeconomics arises because what holds for the individual units may not hold good for the economy as a whole. For example, suppose a firm employs labour for production of output (say, cement). It can hire as many workers it requires at the ongoing wage rate. Thus, increase in demand for labour by a single firm does not have any impact on the wage rate. However, if all the firms in a country increase their demand for labour (say due to economic boom and optimism in the country), there will be shortage of labour and increase in wage rate. Further, the number of workers available for work in the country is limited; thus demand for labour beyond this limit will increase wage rate only, not the supply of labour.

Let us look into the global energy crisis of 2022. This took the form of global shortages and increased the prices of oil, gas and electricity throughout the world. The crisis was caused by a variety of social and economic factors: labour shortages, political disputes, climate concerns, and the war between Russia and Ukraine. The world economy responds to such crises in certain ways. During the global energy crisis of 2022, for example, the oil-exporting countries witnessed an increased inflow of income while petroleum-importing countries experienced a rise in the cost of production. There was severe inflation in most countries. In order to control inflation, policy makers resorted to increase in interest rates and reduction in money supply.

Countries design their monetary and fiscal policies for the welfare of people by controlling inflation and boosting economic growth. Macroeconomic theory guides economists and policymakers to explain the situation and design appropriate policy measures. Among economists, however, there is no agreement on how equilibrium levels of output, prices and employment are determined. Historically economic data are the same, but economists differ on the reasons and solution for economic problems. In microeconomics there is more or less some consensus. In macroeconomics, as you will see, there are sharp and often altogether different explanations and policy recommendations for the same phenomenon.

Macroeconomics has been subjected to debates, particularly since the 1930s. The reasons behind such differences among economists are the basic assumptions and the models they take. Two major schools of thought – Classical and Keynesian – have interpreted economic events differently. Accordingly, they have accorded different explanations for the same economic issue. In this unit, we focus on the classical theories while Keynesian theories will be discussed in Unit 2.

1.2 VARIOUS SCHOOLS OF MACROECONOMIC THOUGHT

A school of macroeconomic thought refers to a group of macroeconomic thinkers who share similar views on the behaviour of macroeconomic variables in an economy. As pointed out above, economists do not always share a similar approach in dealing with macroeconomic problems. For example, Keynesian theory came into existence as a response to the Great Depression. Further economic events during the 1970s led to the revival of the classical views in the form the New Classical Economics. As a response to New Classical economics, a new school of thought – New Keynesian Economics – was born by introducing micro-foundations and rational expectations in Keynesian economics. In later Units of this we will discuss about both these schools of thought.

Classical economics refers to a school of thought prevalent during the eighteenth and early nineteenth century. They believed that the levels of economic variables in an economy were determined through market forces. It implies that wages, prices and interest rates are flexible in an economy. However, the occurrence of the Great Depression (1929-1933), which witnessed massive decline in income, output, employment and price level, a stock market crash, and banking panic falsified such assumptions. This led to the search for new explanations periodic fluctuations in output and employment in an economy. John Maynard Keynes, a British economist, opposed the classical ideas and advocated that prices and wages are not flexible; rather they are sticky. He coined the term "classical" to reflect the ideas presented by economists prior to him. Prominent among classical economists are Adam Smith, David Ricardo, Thomas Malthus, Anne Robert Jacques Turgot, John Stuart Mill, Jean-Baptiste Say, and Eugen Böhm von Bawerk.

As opposed to the classical economists, Keynes advocated that the government should actively intervene in the market to counter recessionary conditions. From Keynes's beliefs evolved a new school of macroeconomic thought that is called the Keynesian school. According to Keynesian economists, actual output could be lower than potential output of an economy and the gap can be closed through fiscal and monetary policy interventions.

There are two major schools of thought: classical and Keynesian. Classical theory emerged in the 18^{th} and 19^{th} centuries. It attempted to explain and analyse the remarkable increase in wealth. Classical theory relies on the assumption of fully flexible prices and wages, which results in an automatic adjustment in the market, ensuring no deficiency of aggregate demand. The economy operates at its full capacity or full employment level. These classical ideas failed to explain and understand the great depression of the 1930s. The depression had affected all the major advanced countries, wherein about one fourth of the labour force had become unemployed. Against this backdrop, John Maynard Keynes came up with his ideas, called Keynesian theory. He attempted to explain the event and advocated that such huge unemployment was because of inadequate aggregate demand, which in turn was the result of a deficiency in investment demand. He criticized the assumption of a *laissez-faire* economy in the classical theory and advocated policy intervention to increase demand and employment.

Later, attempts were made to integrate ideas from both the schools, which led to the "neoclassical synthesis". The IS-LM model that we discuss in Unit 3 is an outcome of the neoclassical synthesis. During the 1950s, economists such as Harrod, Domar and Kaldor extended the Keynesian ideas from short-run to long-run. They argued that there would be an increase in production capacity in the long run because of large investments. This in turn would boost economic growth. The focus of these economists, called the Post-Keynesians, was to understand, explain and analyse economic growth. In order to overcome certain limitations of the post-Keynesian growth models, Robert M. Solow came up with the neoclassical growth model. You will go through these growth models in the course, 'Economics of Growth and Development'.

Keynesian school of thought dominated economic theory and policy after the World War II until the 1970s. During the 1970s most of the advanced economies witnessed stagnation and inflation, a situation called "stagflation". Keynesian theories did not have appropriate policy response to stagflation, which led to the exploration of fresh explanations of economic behavior. Robert E. Lucas, an American economist, revived the classical ideas and countered the prevailing Keynesian thought. The ideas of classical and Keynesian schools differ from each other on various issues, including: i) the role played by demand and supply in determination of equilibrium output, employment and prices; ii) the assumption of flexibility of price level and wage rate in the economy; iii) the dichotomy of real and monetary sectors of the economy, and (iv) neutrality of money.

After the 1970s, however, there was revival of classical economics in the form of new-classical theory. In response to that, new-Keynesian models came up. New Classical economics retained the basic characteristics of classical economics but introduced two new concepts: rational expectations and micro-foundations. We will cover these aspects in later units of this course.

Along with these mainstream schools of thought, there is heterodox schools of thought or heterodox economics. It includes Marxist, Austrian, institutional, feminist and social economics among many others. In fact, there is a need for promoting pluralism in economics these days and researchers are venturing into many new areas. The mainstream or conventional economics, as mentioned, earlier have been Classical and Keynesian. The legacy of these two schools of thought have continued in different forms. Two prominent schools of thought in recent years have been New-Classical economics and New-Keynesian economics. There are certain common features of these two schools of thought: (i) general equilibrium models, (ii) micro-foundations in behaviour of economic agents, and (iii) rational expectations. Two major differences between both the schools of thought are: (i) the assumption regarding market structure – the new-classical school assumes that there is perfect competition in the market while the new-Keynesian school assumes market imperfection. (ii) the assumption of flexibility in wages and prices – the new-classical school assumes flexibility in